

Project Topic: Move & Learn: Motor Skills Development App

Final Project:

The Move & Learn motor skill development app was created through a deeply human centered design process, carefully grounded in the real experiences and needs of children and teens living without eye and hand coordination. Our journey began with thorough needfinding, involving a combination of in person observations, online interactions, interviews with caregivers and therapists, and direct user testing with children facing a variety of motor skill challenges. Each individual's story played a vital role in shaping our design, from Ibrahim's joyful participation to Frohar's need for adaptive pacing, and Emil's desire for engaging, motivating feedback.

Recognizing the diversity among our users was essential. We identified key needs including simplified, gesture-based controls, tools for caregivers to track progress, motivational visual and audio feedback, and adaptable difficulty settings that could flex with the user's abilities. Our heuristic evaluations helped us uncover and address interface issues such as control freedom and system clarity, refining the app to be not only accessible but also genuinely enjoyable and therapeutically effective.

Throughout the design process, the prototype evolved through multiple iterations, informed by continuous feedback, real-world testing, and evidence-based practices. Our team worked in close collaboration, striving to balance technical development with emotional insight and inclusive design principles. We focused on creating interactive activities that are short, intuitive, and rewarding, so users feel supported and empowered rather than pressured. The game encourages repeated physical movement, integrates audio visual rewards, and offers customization options that respond to each child's progress.

The final prototype represents the culmination of our collective efforts, research, and compassion. It transforms theoretical knowledge into a meaningful, engaging, and inclusive digital therapy tool that not only supports motor skill development but also fosters autonomy, joy, and social inclusion for children suffering from eye and hand coordination. We are proud of this work and believe it truly demonstrates the power of thoughtful human computer interaction design when rooted in empathy and real-world understanding.

We are especially thankful to our dear participants whose invaluable contributions made this project possible. Our sincere gratitude extends to our instructor for their continuous feedback and guidance throughout the process, as well as to our classmates who actively took part in the evaluation phase. Most importantly, we thank the children, the real users of the app, for engaging with the game and providing honest feedback that helped us identify and improve design flaws. We are also deeply grateful to the respectful parents, caregivers, therapists, and everyone who participated in the needfinding phase and shared their insights through interviews. Their collective involvement enriched our understanding and strengthened the impact of our work.

Video Prototype:

 **Balloon Game.mov**